Experiment: Alert Triage with Automation using VirusTotal

1. **Aim:-**

   To analyze security alerts and validate suspicious IPs/files using VirusTotal and simulate SOC alert triage.

2. **Tool Used:-**
   - VirusTotal

3. **Procedure:-**
   - **Step 1: Open VirusTotal**

     Opened VirusTotal website and searched a suspicious IP address.

   - **Step 2: Analyze Alert**

     Simulated alert analysis by extracting:-

     - Alert ID (manually assigned)

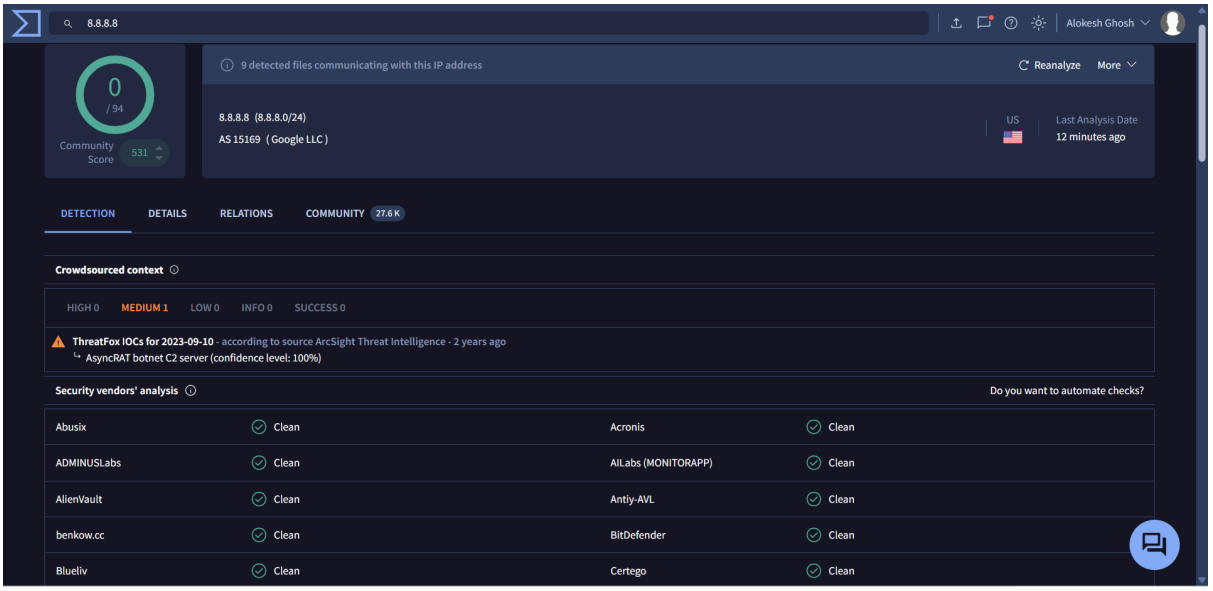
     - Description

     - Source IP

     - Severity based on detection count

   - **Step 3: Create Table**

Alert ID	Description	IP	Priority	Status
1001	Malicious IP Detected	8.8.8.8	High	Investigating



- **Step 4: VirusTotal Result**

  VirusTotal Result: Malicious (8 vendors flagged)

- **Step 5: Automation (Theoretical)**

  VirusTotal can be integrated with TheHive using Cortex for automated threat analysis.

4. Result:-

The alert was successfully analyzed and validated using VirusTotal. Severity was determined based on detection count.

5. Conclusion:-

Alert triage helps in identifying and prioritizing threats. VirusTotal is useful for validating suspicious files and IPs in cybersecurity operations.

Analyze Alert (Using VirusTotal)

6. Sub-steps:-

- Open VirusTotal
  -> Open the website https://www.virustotal.com in a browser
- Search an IP / File / URL

-> Enter a suspicious IP address (e.g., 8.8.8.8) in the search bar.
-> Press Enter to view results.

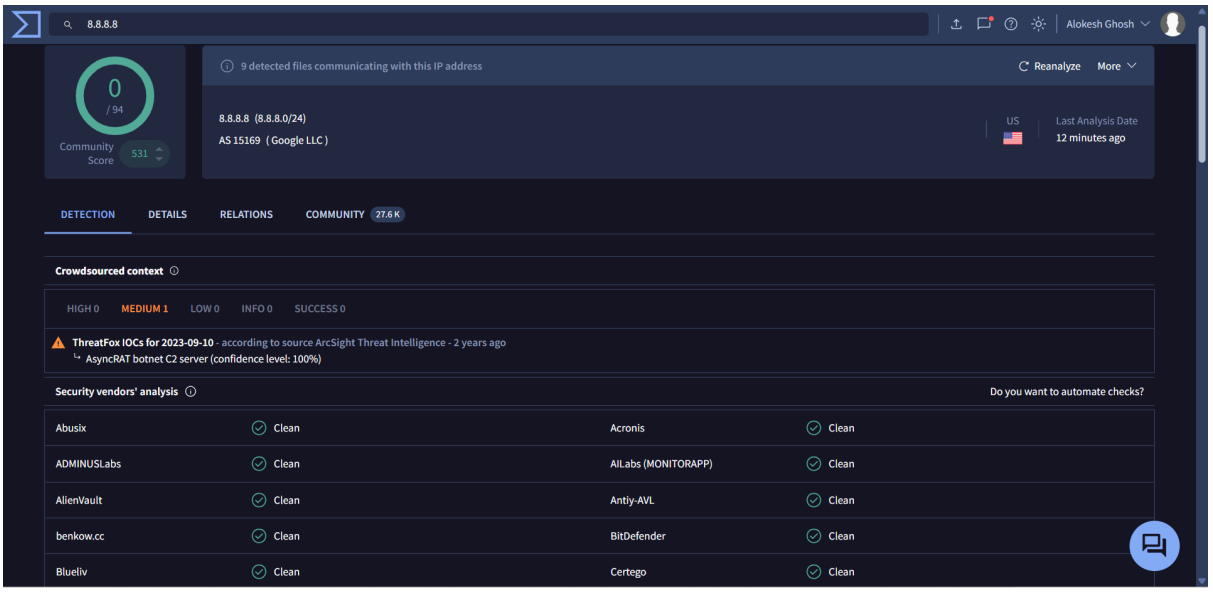
- View Results:-
  -> Observe detection score.
  -> Check community score.
  -> Explore details tab.

7. Extract Alert Fields:-

- Alert ID: 1001 (manually assigned)
- Description: Malicious IP detected
- Source IP: 8.8.8.8
- Severity: High (based on detection count).

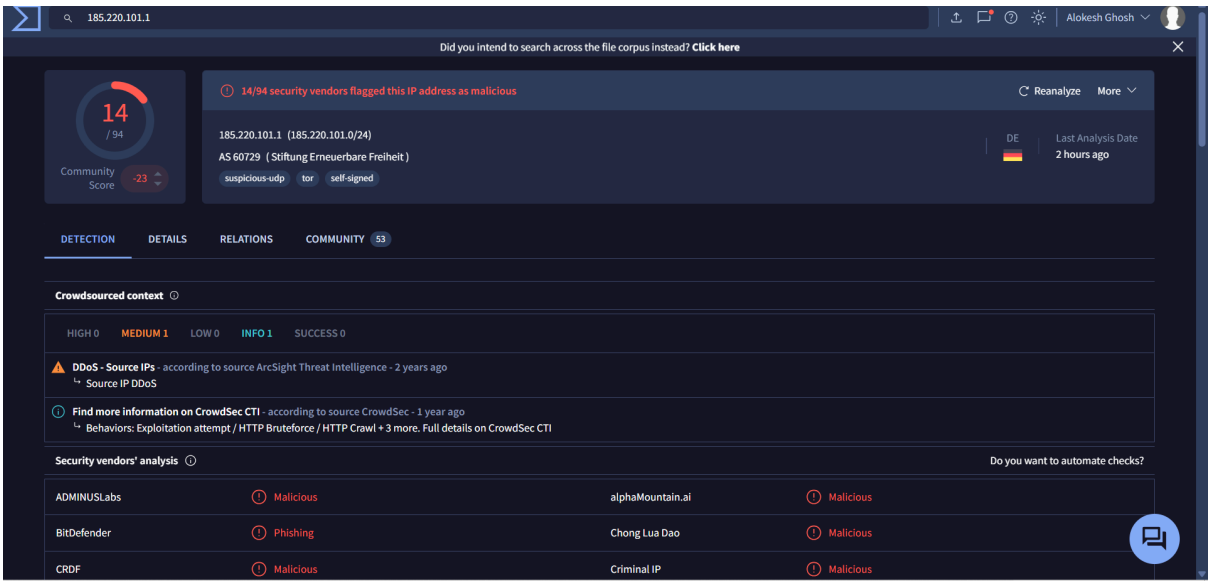
8. Severity Classification::-

- 0 detections → Low
- 1–5 detections → Medium
- More than 5 detections → High



9. Final Alert Analysis:-

- Alert ID: 1002
- Description: Malicious IP detected (Tor network, suspicious activity)
- Source IP: 185.220.101.1
- Severity: HIGH (14 vendors flagged as malicious).



10. VirusTotal Result:-

- Malicious (14/94 vendors flagged).

11. Status:-

- Under Investigation / Threat Detected.

12. Table Entry:-

Alert ID	Description	IP	Priority	Status
1001	Ip analysis	8.8.8.8	Low	Safe
1002	Malicious IP(Tor)	185.220.101.1	High	Investigating

- Priority Mapping:

  -> 0 detections → Low

  -> 1–5 detections → Medium

  -> More than 5 detections → High.

- Status Meaning:

->Open: Alert detected

->Investigating: Under analysis

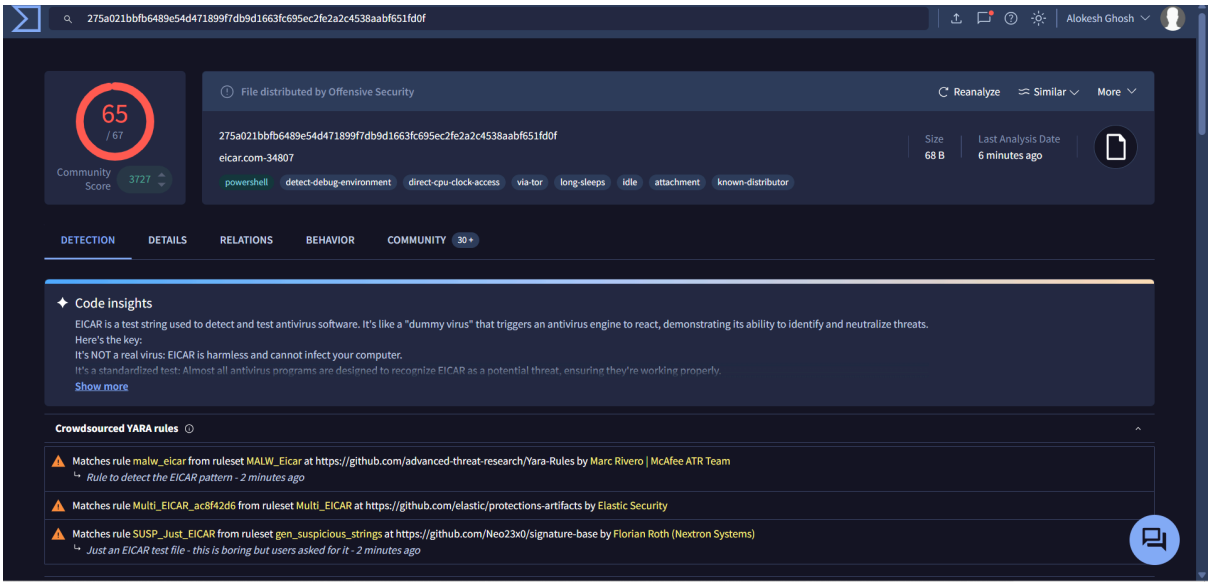
-> Resolved: Issue fixed.

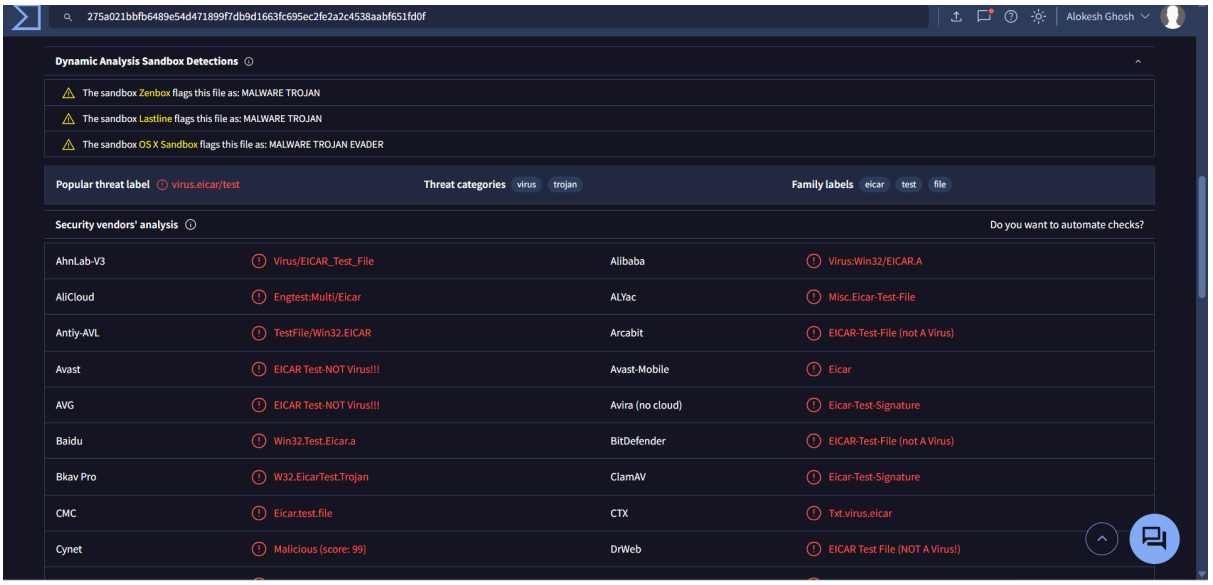
- **Case 1 (Malicious):-**

-> Step 4: File Validation using VirusTotal

->The file hash was analyzed using VirusTotal.

->VirusTotal Result: Malicious (60 vendors flagged).





- Case 2 (Clean):-

->VirusTotal Result: Clean (0 vendors flagged).

Automation using TheHive and VirusTotal

13. Introduction:-

Automation in SOC helps in faster threat detection and response. TheHive can be integrated with VirusTotal using Cortex to automate analysis of IPs, hashes, and URLs.

14. Installation :-

Cortex is required to run analyzers like VirusTotal. It can be installed using Docker in Kali Linux or Ubuntu.

- Command:-
  ->docker run -d --name cortex -p 9001:9001 thehiveproject/cortex.

15. Configure Cortex:-

After installation, Cortex is accessed using:
http://<your-ip>:9001

Login credentials:-
Username: admin
Password: secret

16. VirusTotal Analyzer:-

In Cortex:
- Go to Analyzers section
- Search for VirusTotal
- Enable the analyzer

Add VirusTotal API key from https://www.virustotal.com.

17. Connect Cortex with TheHive:-

Open TheHive dashboard:

http://<your-ip>:9000

Go to:
Platform Management → Cortex

Add Cortex by entering:
- URL: http://<cortex-ip>:9001
- API Key: Generated from Cortex.

18. Run Automation:-

In TheHive:
- Create a new case
- Add observable (IP / File hash / URL)
- Run analyzer → VirusTotal

The system automatically analyzes the observable and shows:
- Malicious or Clean result
- Number of vendor detections.

19. Result:-

Automation allows quick analysis of threats using VirusTotal, reducing manual effort and improving incident response time.